

Project Nginx

Belmouloud Mustapha Abdellah
RSD M1

1/Structure de Reseaux

La 1ère machine (raven@nest) : Elle fait office de point d'entrée. Elle prend en charge le **Load Balancer**, l'**API Gateway**, un **Reverse Proxy** ainsi que la **configuration TLS (HTTPS)**.

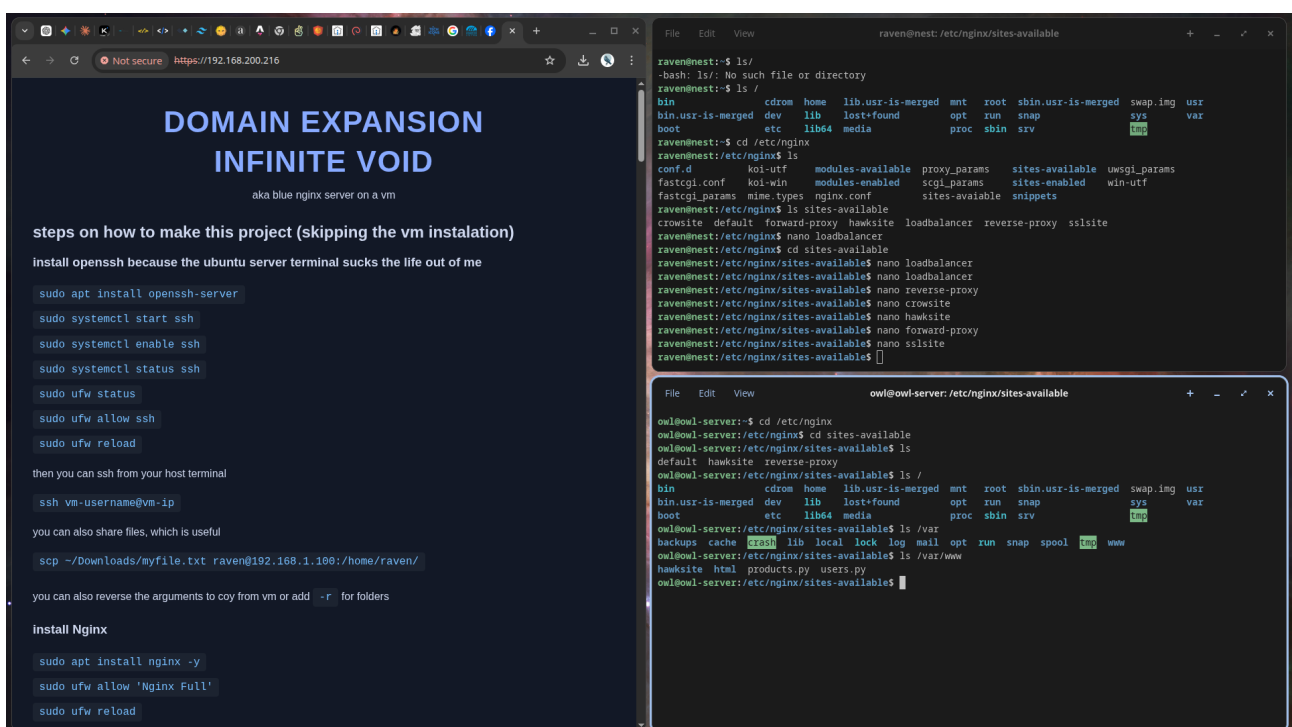
La 2ème machine (owl@owl-server) : Elle héberge les **programmes API** (développés en Python) et un **Reverse Proxy** local.

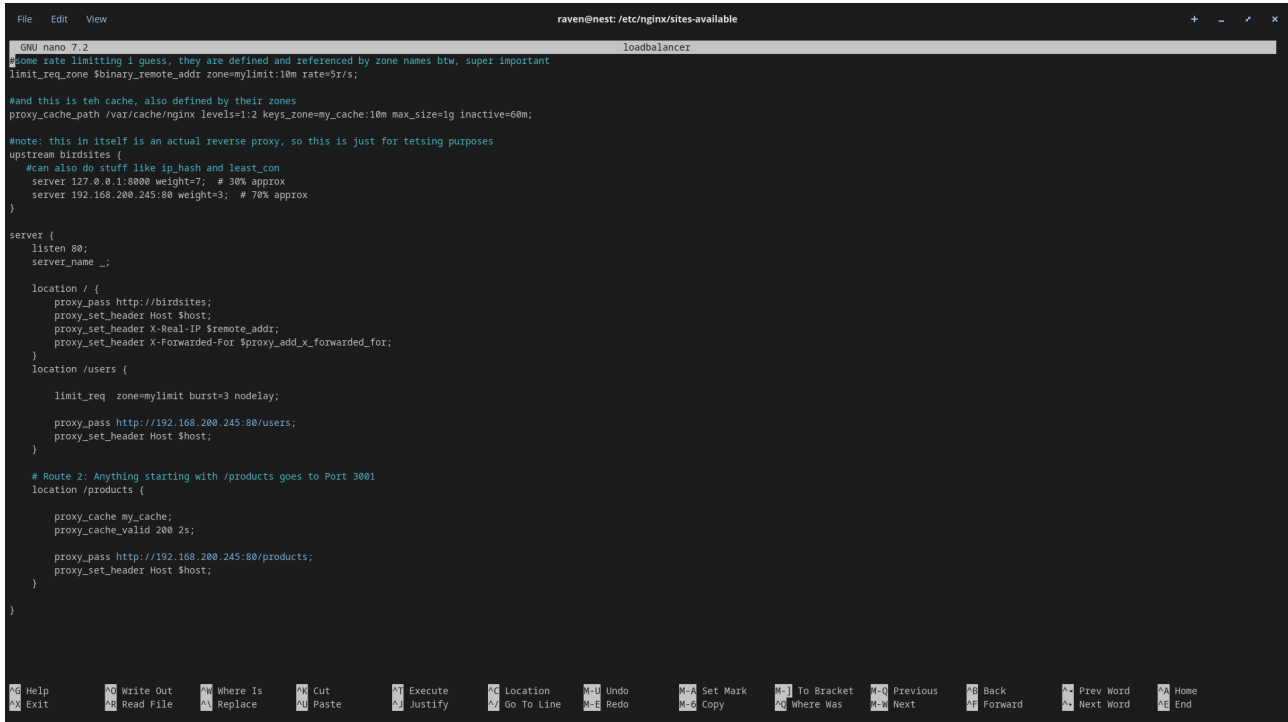
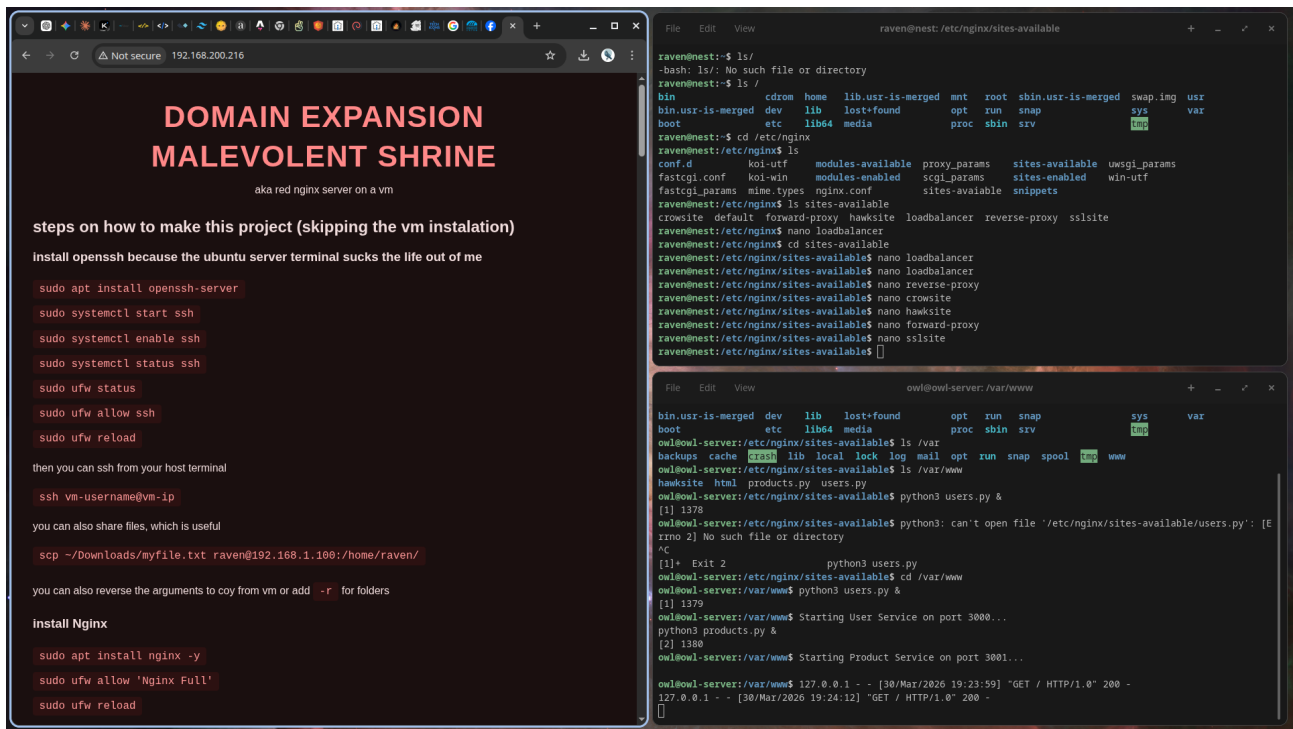
Répartition de charge : La consultation de la page `index.html` est partagée entre les deux machines (équilibrage de charge).

Forward Proxy : Cette fonction a été configurée de deux manières : avec **Squid** et avec **Nginx**. Notez que Nginx n'est pas initialement conçu pour fonctionner comme un *Forward Proxy* natif, d'où l'utilisation de Squid pour une solution plus robuste.

Contenu : La page Web `index.html` présente un résumé personnel détaillant les étapes de configuration suivies.

Captures d'écran des fichiers de configuration:





```
File Edit View raven@nest: /etc/nginx/sites-available
GNU nano 7.2 reverse-proxy
#reverse proxy
server {
    listen 8080;
    server_name _;

    location / {
        proxy_pass http://127.0.0.1:8081;

        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    }

    # Ngix stub_status endpoint
    location /nginx_status {
        stub_status;
        allow 127.0.0.1;      # Only allow localhost by default
        deny all;            # Deny everyone else
    }
}
```

File 'reverse-proxy' is unsuitable

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy To Bracket Where Was Previous Next Back Forward Prev Word Next Word Home End

```
File Edit View raven@nest: /etc/nginx/sites-available
GNU nano 7.2 forward-proxy
server {
    listen 8888;

    resolver 8.8.8.8 ipv6=off;

    allow 192.168.200.0/24;
    deny all;

    access_log /home/raven/forward-proxy.log;

    location / {
        proxy_pass $scheme://$http_host$request_uri;

        proxy_set_header Host $http_host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    }
}
```

File 'forward-proxy' is unsuitable

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy To Bracket Where Was Previous Next Back Forward Prev Word Next Word Home End

```
File Edit View raven@nest:/etc/nginx/sites-available
GNU nano 7.2 sslsite
# --- 1. THE GLOBAL DEFINITIONS (MUST BE OUTSIDE THE SERVER BLOCK) ---
limit_req_zone $binary_remote_addr zone=gatewayssl_limit:10m rate=5r/s;

# This is the line Nginx is complaining about missing:
proxy_cache_path /var/cache/nginxssl levels=1:2 keys=zone-gatewayssl_cache:10m max_size=1g inactive=60m use_temp_path=off;

server {
    listen 443 ssl;
    server_name _;

    # SSL Files
    ssl_certificate /etc/ssl/certs/nginx-selfsigned.crt;
    ssl_certificate_key /etc/ssl/private/nginx-selfsigned.key;

    # Helpful Header to see if Cache is working
    add_header X-Cache-Status $upstream_cache_status;

    location / {
        # Assuming your site is in /var/www/crowsite
        root /var/www/hawksite;
        index index.html;
        try_files $uri $uri/ =404;
    }

    # GATEWAY ROUTE: Users
    location /users {

        limit_req zone=gatewayssl_limit burst=10 nodelay; # Rate Limit

        proxy_pass http://192.168.200.245:3000/;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
    }

    # GATEWAY ROUTE: Products (With Caching)
    location /products {

        proxy_cache gatewayssl_cache; # Enable Caching
        proxy_cache_valid 200 1m; # Cache success for 1 min

        proxy_pass http://192.168.200.245:3001/;
        proxy_set_header Host $host;
    }
}
```

```
File Edit View raven@nest:/etc/nginx/sites-available
raven@nest:~$ ls/
-bash: ls/: No such file or directory
raven@nest:~$ ls /
bin          cdrom        home         lib.usr-is-merged  mnt         root        sbin.usr-is-merged  swap.img  usr
bin.usr-is-merged  dev         lib          lost+found         opt         run         snap                sys       var
boot         etc          lib64        media              proc        sbin        srv                 tmp
raven@nest:~$ cd /etc/nginx
raven@nest:/etc/nginx$ ls
conf.d      koi-utf      modules-available  proxy_params  sites-available  uwsgi_params
fastcgi.conf  koi-win      modules-enabled    scgi_params   sites-enabled    win-utf
fastcgi_params  mime.types  nginx.conf        sites-available  snippets
raven@nest:/etc/nginx$ ls sites-available
crowsite default forward-proxy hawksite loadbalancer reverse-proxy sslsite
raven@nest:/etc/nginx$ nano loadbalancer
raven@nest:/etc/nginx$ cd sites-available
raven@nest:/etc/nginx/sites-available$ nano loadbalancer
raven@nest:/etc/nginx/sites-available$ nano reverse-proxy
raven@nest:/etc/nginx/sites-available$ nano crowsite
raven@nest:/etc/nginx/sites-available$ nano hawksite
raven@nest:/etc/nginx/sites-available$ nano forward-proxy
raven@nest:/etc/nginx/sites-available$ nano sslsite
raven@nest:/etc/nginx/sites-available$

File Edit View owl@owl-server:/var/www
bin.usr-is-merged dev lib lost+found opt run snap sys var
boot etc lib64 media proc sbin srv tmp
owl@owl-server:/etc/nginx/sites-available$ ls /var
backups cache tmp lib local lock log mail opt run snap spool tmp www
owl@owl-server:/etc/nginx/sites-available$ ls /var/www
hawksite html products.py users.py
owl@owl-server:/etc/nginx/sites-available$ python3 users.py &
[1] 1378
owl@owl-server:/etc/nginx/sites-available$ python3: can't open file '/etc/nginx/sites-available/users.py': [Errno 2] No such file or directory
^C
[1]+ Exit 2 python3 users.py
owl@owl-server:/etc/nginx/sites-available$ cd /var/www
owl@owl-server:/var/www$ python3 users.py &
[1] 1379
owl@owl-server:/var/www$ Starting User Service on port 3000...
python3 products.py &
[2] 1380
owl@owl-server:/var/www$ Starting Product Service on port 3001...

owl@owl-server:/var/www$ 127.0.0.1 - - [30/Mar/2026 19:23:59] "GET / HTTP/1.0" 200 -
127.0.0.1 - - [30/Mar/2026 19:24:12] "GET / HTTP/1.0" 200 -
^
```

```

192.168.200.216/users
{"service": "User Service", "port": 3000, "status": "active"}

raven@nest:~$ ls
-bash: ls: No such file or directory
raven@nest:~$ ls /
bin          cdrom        home         lib.usr-is-merged  mnt          root        sbin.usr-is-merged  swap.img  usr
bin.usr-is-merged  dev         lib          lost+found        opt          run         snap              sys       var
boot         etc          lib64        media             proc         sbin        srv               [top]
raven@nest:~$ cd /etc/nginx
raven@nest:/etc/nginx$ ls
confd          koi-utf      modules-available  proxy_params  sites-available  uwsgi_params
fastcgi.conf  koi-win     modules-enabled    scgi_params   sites-enabled    win-utf
fastcgi_params  mime.types  nginx.conf        sites-available  snippets
raven@nest:/etc/nginx$ ls sites-available
crowdsec default forward-proxy hawksite loadbalancer reverse-proxy sslsite
raven@nest:/etc/nginx$ nano loadbalancer
raven@nest:/etc/nginx$ cd sites-available
raven@nest:/etc/nginx/sites-available$ nano loadbalancer
raven@nest:/etc/nginx/sites-available$ nano loadbalancer
raven@nest:/etc/nginx/sites-available$ nano reverse-proxy
raven@nest:/etc/nginx/sites-available$ nano crowdsec
raven@nest:/etc/nginx/sites-available$ nano hawksite
raven@nest:/etc/nginx/sites-available$ nano forward-proxy
raven@nest:/etc/nginx/sites-available$ nano sslsite
raven@nest:/etc/nginx/sites-available$

owl@owl-server:/var/www$ ls
bin          cdrom        home         lib.usr-is-merged  mnt          root        sbin.usr-is-merged  swap.img  usr
bin.usr-is-merged  dev         lib          lost+found        opt          run         snap              sys       var
boot         etc          lib64        media             proc         sbin        srv               [top]
owl@owl-server:/etc/nginx/sites-available$ ls /var
backups  cache  [top]  lib  local  lock  log  mail  opt  run  snap  spool  [top]  www
owl@owl-server:/etc/nginx/sites-available$ ls /var/www
hawksite  html  products.py  users.py
owl@owl-server:/etc/nginx/sites-available$ python3 users.py &
[1] 1378
owl@owl-server:/etc/nginx/sites-available$ python3: can't open file '/etc/nginx/sites-available/users.py': [Errno 2] No such file or directory
^C
[1]+  Exit 2                  python3 users.py
owl@owl-server:/etc/nginx/sites-available$ cd /var/www
owl@owl-server:/var/www$ python3 users.py &
[1] 1379
owl@owl-server:/var/www$ Starting User Service on port 3000...
python3 products.py &
[2] 1380
owl@owl-server:/var/www$ Starting Product Service on port 3001...

owl@owl-server:/var/www$ 127.0.0.1 - - [30/Mar/2026 19:23:59] "GET / HTTP/1.0" 200 -

```

